Project 2: 2D Finite Element Analysis of Microstrip Transmission Lines for High-Speed PCB Signal Integrity

Numerical Analysis of Differential Equations

1. Background and Motivation

In modern high-speed digital design and RF engineering, *Signal Integrity (SI)* is paramount. A printed circuit board (PCB) trace must be treated as a **transmission line** when its electrical length is not negligible compared with the shortest significant wavelength of the signal. For digital signals this criterion is usually governed by edge rise time, not by a universal clock-frequency threshold. If the *characteristic impedance* (Z_0) of the trace is not adequately matched to the source and load, signal reflections can occur and degrade signal quality.

The characteristic impedance is determined by the line's per-unit-length inductance and capacitance, which in turn depend on cross-sectional geometry and material properties. This project focuses on the widely used **microstrip line**. Your goal is to model the 2D cross-section, solve the governing electrostatic boundary-value problem (BVP) using the **Finite Element Method (FEM)**, and rigorously analyze the numerical challenges introduced by material interfaces and geometric singularities.

2. Mathematical Model

Let Ω_R denote the truncated air–substrate field region with the perfectly conducting trace interior removed. Thus the trace is an internal boundary of Ω_R, not a material subdomain in which the electrostatic PDE is solved. The electrostatic potential $\phi(x, y)$ is governed by the generalized Laplace equation:

$$-\nabla \cdot (\epsilon(x, y)\nabla\phi) = 0, \quad (x, y) \in \Omega_R \tag{1}$$

where $\epsilon(x, y) = \epsilon_r(x, y)\epsilon_0$ is the spatially varying permittivity of the medium.

2.1 Boundary and Interface Conditions

- **Dirichlet BCs:** $\phi = V_0$ on the signal trace boundary Γ_{trace}, and $\phi = 0$ on the ground plane Γ_{ground}.
- **Material Interfaces:** Across the air–substrate interface outside the trace footprint, $[\phi] = 0$ and $[\epsilon\nabla\phi\cdot n] = 0$, where one common orientation of n is used on both sides. Equivalently, if n_i is the outward normal of subdomain i, then $\epsilon_1\partial_{n_1}\phi_1 + \epsilon_2\partial_{n_2}\phi_2 = 0$.
- **Open Boundary Truncation:** The physical domain extends to infinity. We truncate it to a finite bounding box Ω_R and impose a stated artificial condition on Γ_{far}, such as homogeneous Neumann $\epsilon\nabla\phi \cdot n = 0$ or homogeneous Dirichlet $\phi = 0$. These alternatives have different truncation biases and must not be mixed within one convergence study.

2.2 Weak Formulation

The essential boundary and its prescribed trace depend on the selected far-field truncation:

far-field choice	Γ_D	g_D
$\epsilon\nabla\phi \cdot n = 0$ on Γ_{far}	$\Gamma_{\text{trace}} \cup \Gamma_{\text{ground}}$	V_0 on Γ_{trace}, 0 on Γ_{ground}
$\phi = 0$ on Γ_{far}	$\Gamma_{\text{trace}} \cup \Gamma_{\text{ground}} \cup \Gamma_{\text{far}}$	V_0 on Γ_{trace}, 0 on $\Gamma_{\text{ground}} \cup \Gamma_{\text{far}}$.

For the chosen row, define the affine trial space and homogeneous test space

$$\mathcal{V}_g = \{u \in H^1(\Omega_R) : u = g_D \text{ on } \Gamma_D\},$$
$$\mathcal{V}_{00} = \{v \in H^1(\Omega_R) : v = 0 \text{ on } \Gamma_D\}.$$

The weak problem for either choice is: find $\phi \in \mathcal{V}_g$ such that for every $v \in \mathcal{V}_{00}$,

$$\int_{\Omega_R} \epsilon(x,y)\nabla\phi \cdot \nabla v \, dxdy = 0 \tag{2}$$

because the boundary term is zero either through the homogeneous natural Neumann condition or because the test function vanishes on the Dirichlet far boundary. Choose one row and use it consistently in the domain-growth study. *Note: For a conforming H^1 discretization with the discontinuous coefficient ϵ integrated correctly, the weak form enforces interface flux continuity in the weak (distributional) sense without Lagrange multipliers. It does not imply pointwise equality of the two discrete normal-flux traces.*

Once ϕ is solved, the capacitance per unit length C is computed via the electrostatic energy or Gauss's law, and the characteristic impedance is $Z_0 = \frac{\sqrt{\epsilon_{\text{eff}}}}{c_0 C}$, where c_0 is the speed of light in vacuum.

3. Geometric and Physical Parameters

You will model an idealized microstrip cross-section whose nominal impedance is near 50Ω. All dimensions are in millimeters (mm).

- **Ground Plane:** Located at $y = 0$.
- **Dielectric Substrate (idealized FR-4):** Region $0 < y < H$, where $H = 1.6$ mm. Use a lossless, isotropic nominal relative permittivity $\epsilon_r = 4.4$. Real FR-4 is frequency-dependent, anisotropic, and process-dependent.
- **Air Region:** Region $y > H$. Relative permittivity $\epsilon_r = 1.0$.
- **Signal Trace (perfect conductor):** Exclude the rectangular trace interior from Ω_R and impose $\phi = V_0$ on its boundary. It is centered at $x = 0$, occupies $H \le y \le H + t$, has width $w = 3.0$ mm, and thickness $t = 0.035$ mm (nominal 1 oz copper).
- **Computational Domain Ω_R:** $x \in [-W, W]$, $y \in [0, H_{\text{air}}]$. (You must determine appropriate values for W and H_{air}).

4. Project Objectives and Tasks

This is a closed-loop project: Physical Geometry → PDE → FEM → Numerical Analysis → Engineering Validation.

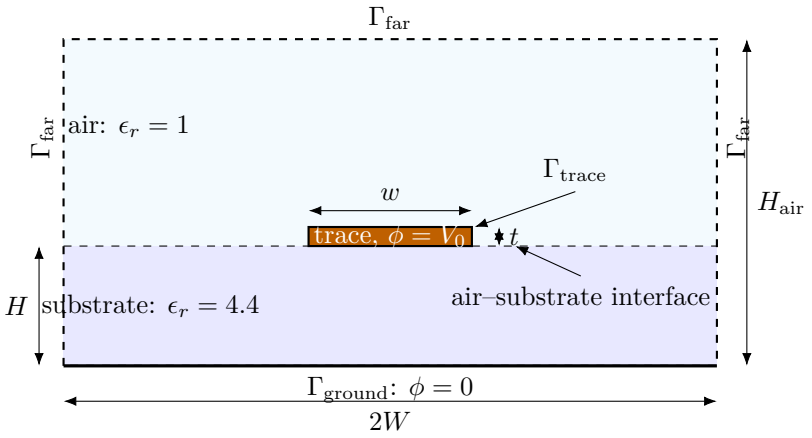


Figure 1: Truncated microstrip cross-section (not to scale). The field domain Ω_R is the air–substrate rectangle with the conductor interior removed. The artificial boundary comprises the top and two vertical sides.

Task 1: Mathematical Derivation: Derive the weak formulation. Explain why a conforming FEM with correctly represented discontinuous permittivity enforces $[\epsilon \nabla \phi \cdot n] = 0$ in the weak sense without Lagrange multipliers, while not generally making the two discrete normal-flux traces pointwise identical.

Task 2: FEM Implementation: Implement the solver (e.g., using FEniCS, FreeFEM++, or MATLAB PDE Toolbox). **AI Usage Policy:** You may use AI to generate boilerplate meshing or assembly code. However, you *must* manually define the subdomains for $\epsilon(x, y)$ and verify the boundary conditions. Plot ϕ_h, $|\mathbf{E}_h|$, and the elementwise energy density. Check $0 \leq \phi_h \leq V_0$, symmetry about $x = 0$, matrix symmetry and positive definiteness after Dirichlet elimination, and global charge balance.

Task 3: Domain Truncation Analysis (Learning Outcome D): Choose one artificial boundary condition and use

$$(W/H,\, H_{\text{air}}/H) = (3, 3), (5, 5), (8, 8), (12, 12).$$

Preserve the same near-trace mesh on every domain and extend only the outer mesh. First make the fixed-domain discretization error demonstrably smaller than the intended truncation differences. Use the $(12, 12)$ result as the stated truncation reference and report $|C_R - C_{12}|/|C_{12}|$. Determine the smallest tested domain below 1%, then repeat the largest domain with finer near-field resolution to check that the conclusion is not a mesh artifact. Homogeneous Dirichlet models an enclosing grounded box and typically adds capacitance; homogeneous Neumann suppresses normal far-boundary flux. Check the observed bias and discuss cost. Do not mix the conditions in one sequence.

Task 4: Corner Singularity and Convergence (Learning Outcome E - *Core Task*): In the field domain, a convex conductor corner appears as a re-entrant boundary corner. The electric field $\mathbf{E} = -\nabla\phi$ can therefore be singular there; at the lower trace corners the conductor also meets the air–substrate interface, so the singular exponent depends on geometry and permittivity contrast.

- Perform a mesh refinement study using a *uniform* mesh. Use the energy norm

$$\|e\|_E^2 = \int_{\Omega_R} \epsilon |\nabla e|^2 \, d\Omega,$$

and replace the unavailable exact solution by a substantially over-refined reference ϕ_{ref}. Evaluate $\nabla(\phi_{\text{ref}} - \phi_h)$ on the reference mesh after interpolating or projecting ϕ_h into that space, or integrate on an explicitly constructed common overlay; never subtract coefficient vectors from different meshes. Demonstrate that further reference refinement changes the reported errors negligibly. Plot the relative energy-norm error against the quasi-uniform mesh size h and estimate the observed rate.

- **Mitigation Strategy:** Design a *graded mesh* or a posteriori-driven *local adaptive refinement* strategy that concentrates elements near the trace corners and conductor–material junctions. A single global h is not meaningful on a locally refined mesh: report error against degrees of freedom, nonzeros, and solve time, and compare with the uniform sequence at matched work. State the estimator and marking rule. Determine whether smooth-solution efficiency for the chosen element order is recovered; do not assume that recovery is automatic.

Task 5: Real-World Validation: Using your most accurate FEM result, calculate Z_0. Treat Hammerstad–Jensen as the principal closed-form benchmark and IPC-2141 as a coarse engineering cross-check. Discuss discrepancies in terms of formula approximation, finite thickness, truncation, discretization, and material idealization rather than treating either formula as exact. Benchmark agreement is a model-level check: it does not replace the mesh-refinement evidence of Task 4.

Appendix: Characteristic Impedance of Microstrip Lines
Formulas and Extraction from FEM

A Quasi-TEM Model of Microstrip Lines

A microstrip line supports a quasi-TEM mode. Under this lossless, quasistatic approximation, the per-unit-length inductance L and capacitance C fully characterize the line. The two fundamental relations are:

$$Z_0 = \sqrt{\frac{L}{C}}, \qquad v_p = \frac{1}{\sqrt{LC}} = \frac{c_0}{\sqrt{\varepsilon_{\text{eff}}}} \tag{3}$$

where v_p is the phase velocity, $c_0 = 1/\sqrt{\mu_0 \varepsilon_0}$ is the speed of light in vacuum, and ε_{eff} is the effective relative permittivity.

This model does not predict frequency dispersion, dielectric or conductor loss, copper roughness, solder-mask loading, radiation, anisotropy, or fabrication tolerances. The extracted Z_0 is therefore a quasistatic nominal value, not a broadband signal-integrity model. Important omitted effects require frequency-dependent material data and, when necessary, a full-wave or lossy transmission-line analysis.

Within the lossless quasistatic model and for non-magnetic media ($\mu = \mu_0$), the external inductance is determined by geometry rather than dielectric permittivity. If the dielectric is replaced by air ($\varepsilon_r = 1$) while preserving the conductor geometry and truncation treatment, the corresponding TEM relation gives:

$$L \cdot C_{\text{air}} = \frac{1}{c_0^2} \quad \Longrightarrow \quad L = \frac{1}{c_0^2 \, C_{\text{air}}} \tag{4}$$

B Extraction of C, ε_{eff}, and Z_0 from FEM

B.1 Step 1: Compute C (with dielectric)

Solve the 2D Laplace equation with the idealized dielectric ($\varepsilon_r = 4.4$ for nominal FR-4):

$$-\nabla \cdot \big(\varepsilon(x, y)\, \nabla \phi\big) = 0 \tag{5}$$

with $\phi = V_0$ on the trace and $\phi = 0$ on the ground plane. The capacitance per unit length is:

$$C = \frac{|Q|}{|V_0|}, \qquad Q = \oint_{\Gamma_{\text{trace}}} \varepsilon \, \frac{\partial \phi}{\partial n_\Omega} \, ds \tag{6}$$

where n_Ω points outward from the dielectric field domain and into the conductor. If the conductor-outward normal is used instead, the charge integral acquires a minus sign. In FEM, compute an independent energy value

$$C_E = \frac{2\, W_e}{V_0^2}, \qquad W_e = \frac{1}{2} \int_\Omega \varepsilon \, |\nabla \phi|^2 \, dx \, dy, \qquad C_Q = \frac{|Q|}{|V_0|}. \tag{7}$$

Raw differentiation of a low-order finite-element solution on Γ_{trace} is sensitive to the corner singularities. Obtain Q from the assembled Dirichlet reaction, an equilibrated/recovered flux, or a documented weak contour method rather than blindly integrating an elementwise raw gradient. Report

$$\delta_{EQ} = \frac{|C_E - C_Q|}{C_E};$$

What δ_{EQ} is, and what it is not. When Q comes from the assembled Dirichlet reaction, C_E and C_Q are tied by an *algebraic identity* of the same stiffness matrix K. Let u be the full solved coefficient vector (interior values from the solve, prescribed values $V_0/0$ on the trace/ground dofs). With no volume load the energy is $W_e = \frac{1}{2}u^\top Ku$, and the reaction is Ku restricted to the Dirichlet dofs; the interior rows of Ku vanish, so $2W_e = u^\top Ku = V_0 \sum_{\Gamma_{\text{trace}}} (Ku)_i = V_0 Q$ up to solver tolerance, on any mesh — coarse included. δ_{EQ} is therefore a *consistency check*: it confirms C_Q was extracted from the correct dof, sign, and quadrature, *not* that the field has converged. Spatial convergence is established only by mesh refinement: report C_E (and C_Q if reported) on a sequence of meshes approaching a limiting value — compared against an over-refined reference ϕ_{ref} on a finer mesh, *never* the same mesh. A closed-form benchmark such as Hammerstad–Jensen (Task 5) checks the physical model and the extracted definitions, not the discretisation, and cannot replace mesh refinement. Report C_E, C_Q, δ_{EQ}, and the refinement study together; δ_{EQ} alone, or closed-form agreement alone, does not establish spatial convergence. Use the converged energy extraction as $C := C_E$ in the formulas below; report C_Q through δ_{EQ} only — it is a consistency check, not an independent validation.

B.2 Step 2: Compute C_{air} (air only)

Repeat the solve with $\varepsilon_r = 1$ everywhere. Preserve the conductor geometry, computational domain, mesh, element degree, quadrature, solver tolerance, and artificial boundary condition exactly. Only the permittivity field changes. Extract C_{air} by the same energy procedure used for C, and repeat the energy–charge consistency check. Unmatched numerical settings contaminate the ratio below.

B.3 Step 3: Effective permittivity

$$\boxed{\varepsilon_{\text{eff}} = \frac{C}{C_{\text{air}}}} \tag{8}$$

Physically, ε_{eff} satisfies $1 < \varepsilon_{\text{eff}} < \varepsilon_r$ since the field lines are partly in air and partly in the substrate.

B.4 Step 4: Characteristic impedance

Combining $Z_0 = \sqrt{L/C}$ with $L = 1/(c_0^2 C_{\text{air}})$:

$$\boxed{Z_0 = \frac{1}{c_0\sqrt{C \cdot C_{\text{air}}}} = \frac{\sqrt{\varepsilon_{\text{eff}}}}{c_0 \cdot C}} \tag{9}$$

Practical Note

In FEM practice, C and C_{air} must have units of F/m. Convert all geometry to metres before assembly, or convert the extracted per-unit-length quantity consistently before computing Z_0 in ohms. For the idealized parameters here, the expected quasistatic result is near $50\,\Omega$. Report C_E, C_Q, C_{air}, δ_{EQ}, ε_{eff}, and Z_0 with units and enough digits to expose convergence.

C IPC-2141 Empirical Formula

The following compact empirical approximation is commonly associated with the IPC-2141 design guidance [1] for an **external microstrip** (one reference plane, trace on the top surface):

$$\boxed{Z_0^{\mathrm{IPC}} = \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln\!\left(\frac{5.98\,H}{0.8\,w + t}\right)} \tag{10}$$

where:

Symbol	Meaning	Unit
H	Dielectric substrate thickness	mm
w	Trace width	mm
t	Trace (copper) thickness	mm
ε_r	Relative permittivity of substrate	—

This compact expression is an engineering approximation, not an exact field solution. Use it only as a coarse cross-check, keep all lengths in the same unit, and document the source and applicability limits adopted for any graded work. The present geometry has $w/H = 1.875$ and $t/H \approx 0.0219$.

Remark 1. *For the project parameters ($w = 3.0$, $H = 1.6$, $t = 0.035$, $\varepsilon_r = 4.4$):*

$$\begin{aligned} Z_0^{IPC} &= \frac{87}{\sqrt{4.4 + 1.41}} \ln\!\left(\frac{5.98 \times 1.6}{0.8 \times 3.0 + 0.035}\right) \\ &= \frac{87}{\sqrt{5.81}} \ln\!\left(\frac{9.568}{2.435}\right) \\ &= \frac{87}{2.411} \times \ln(3.929) \\ &= 36.09 \times 1.368 \\ &\approx 49.4\ \Omega \end{aligned}$$

This is close to the target 50 Ω, confirming the geometry is reasonable.

D Hammerstad–Jensen Quasistatic Model

For a more accurate closed-form reference, use the Hammerstad–Jensen model [2]. The following equations include its finite strip-thickness correction. They are quasistatic: frequency dispersion, conductor roughness, dielectric loss, and fabrication tolerances are not included.

Set the normalized width and thickness to

$$u = \frac{w}{H}, \qquad \tau = \frac{t}{H}. \tag{11}$$

For any positive normalized width s, define

$$a(s) = 1 + \frac{1}{49}\ln\!\left(\frac{s^4 + (s/52)^2}{s^4 + 0.432}\right) + \frac{1}{18.7}\ln\!\left(1 + (s/18.1)^3\right), \tag{12}$$

$$b(\varepsilon_r) = 0.564\left(\frac{\varepsilon_r - 0.9}{\varepsilon_r + 3}\right)^{0.053}, \tag{13}$$

$$\varepsilon_{\mathrm{HJ}}(s) = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{10}{s}\right)^{-a(s)b(\varepsilon_r)}, \tag{14}$$

$$f(s) = 6 + (2\pi - 6)\exp\!\left[-\left(\frac{30.666}{s}\right)^{0.7528}\right], \tag{15}$$

$$Z_{\mathrm{air}}(s) = 60\ln\!\left[\frac{f(s)}{s} + \sqrt{1 + \left(\frac{2}{s}\right)^2}\right]\ \Omega. \tag{16}$$

For an infinitesimally thin strip, $\varepsilon_{\mathrm{eff}} = \varepsilon_{\mathrm{HJ}}(u)$ and $Z_0 = Z_{\mathrm{air}}(u)/\sqrt{\varepsilon_{\mathrm{eff}}}$.

For $t > 0$, the Hammerstad–Jensen effective-width corrections are

$$\Delta u_1 = \frac{\tau}{\pi}\ln\!\left[1 + \frac{4e}{\tau}\tanh^2\!\left(\sqrt{6.517u}\right)\right], \tag{17}$$

$$\Delta u_r = \frac{\Delta u_1}{2}\left[1 + \operatorname{sech}\!\left(\sqrt{\varepsilon_r - 1}\right)\right], \tag{18}$$

$$u_1 = u + \Delta u_1, \qquad u_r = u + \Delta u_r. \tag{19}$$

The thickness-corrected quasistatic results can then be written in either equivalent form

$$\varepsilon_{\mathrm{eff},t} = \varepsilon_{\mathrm{HJ}}(u_r)\left[\frac{Z_{\mathrm{air}}(u_1)}{Z_{\mathrm{air}}(u_r)}\right]^2, \tag{20}$$

$$Z_{0,t} = \frac{Z_{\mathrm{air}}(u_r)}{\sqrt{\varepsilon_{\mathrm{HJ}}(u_r)}} = \frac{Z_{\mathrm{air}}(u_1)}{\sqrt{\varepsilon_{\mathrm{eff},t}}}. \tag{21}$$

For $w = 3.0$ mm, $H = 1.6$ mm, $t = 0.035$ mm, and $\varepsilon_r = 4.4$,

$$\begin{aligned} u &= 1.875, & \tau &= 0.021875,\\ \Delta u_1 &= 0.043220, & \Delta u_r &= 0.028280,\\ u_1 &= 1.918220, & u_r &= 1.903280,\\ \varepsilon_{\mathrm{eff},t} &\approx 3.3008, & Z_{0,t} &\approx 50.20\ \Omega. \end{aligned}$$

E Comparison Table for Student Validation

Students should compare their FEM-extracted Z_0 against both formulas. Use Hammerstad–Jensen as the named denominator and report the relative *discrepancy*

$$d_{\mathrm{HJ}}(Z) = \frac{|Z - Z_{0,t}^{\mathrm{HJ}}|}{Z_{0,t}^{\mathrm{HJ}}}.$$

Method	Z_0 (Ω)	d_{HJ}
FEM (your result)	—	—
IPC-2141	≈ 49.4	$\approx 1.6\%$
Hammerstad–Jensen (finite t)	≈ 50.20	0 (reference)

The closed-form results are useful cross-checks, not exact acceptance bounds; d_{HJ} is a discrepancy, not a known FEM error. A converged FEM result must use matched geometry and material assumptions. Discuss remaining differences in terms of formula approximation, finite thickness, truncation, material idealization, and discretization error.

References

[1] IPC-2141, *Controlled Impedance Circuit Boards and High Speed Logic Design*, IPC, April 1996. Superseded by IPC-2141A, *Design Guide for High-Speed Controlled Impedance Circuit Boards*, March 2004.

[2] E. Hammerstad and O. Jensen, "Accurate models for microstrip computer-aided design," in *1980 IEEE MTT-S International Microwave Symposium Digest*, 1980, pp. 407–409, doi: 10.1109/MWSYM.1980.1124303.

[3] D. M. Pozar, *Microwave Engineering*, 4th ed. Wiley, 2011. (Chapter 3: Transmission Lines and Microstrip)